Hanxiang Shen

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EDA R&D Engineer at HiSilicon with a PhD in Computational Mathematics from Fudan University. Working on the design and C++ implementation of numerical algorithms for EDA tools, with a focus on timing analysis and power optimization.

EDUCATION

PhD, Computational Mathematics 2019.09 – 2024.07
Fudan University · Shanghai Center for Mathematical Sciences
Research: Numerical algorithms in computational chemistry, EDA algorithms.

BSc, Information and Computational Science 2015.09 – 2019.07
Fudan University · School of Mathematical Sciences

EXPERIENCE

R&D Engineer 2024.07 – Present
HiSilicon (Huawei) · Chip Development Platform

- Engaged in EDA simulation algorithm R&D for physical design, focusing on timing and power domains.
- Primary tech stack: C++ and Tcl. Proficient with STL; capable of independent module-level development and Tcl automation scripting.
- Project 1: Contributed to the SI module algorithm R&D in the in-house timing engine, modeling crosstalk noise and delay in interconnects. Responsible for prototype validation and production deployment of this module, supporting timing analysis accuracy improvements. Gained in-depth understanding of benchmark tool crosstalk modules and SPICE deck tools, developed multiple automated verification scripts.
- Project 2: Contributed to the in-house Glitch Power optimization tool development, responsible for the digital waveform simulation module and core ECO optimization strategy development. Built a solution integrating VCS+PT+Innovus+in-house tools. Achieved power savings in AI core designs.

R&D Intern 2023.06 – 2023.08
Hongxin Microelectronics

- Developed timing check module backend in the STA tool (C++), implementing constraint propagation and violation reporting across multiple clock domains.

Algorithm R&D Intern 2023.02 – 2023.05
HiSilicon (Huawei)

- Constructed mathematical models for a 7nm-node EDA feature and reverse-engineered benchmark tool behavior to validate accuracy; delivered production-quality implementation matching benchmark results.

AWARDS

3rd Prize — 1st GBA (Huangpu) International Algorithm Competition — SVD and Inversion of Approximately Low-Rank Matrices 2022.11

2nd Prize — 2nd Pioneer Cup Computational Application Award — Quantum Computing 2021.10

Honorable Mention — Mathematical Contest in Modeling (MCM) 2017.01

SKILLS

Programming	C++, Python, MATLAB
EDA Domain	Static Timing Analysis, crosstalk (SI), power analysis, SPICE-level simulation, Liberty/SDC
Mathematics	Numerical linear algebra, optimization, ODE/PDE solvers, mathematical modeling

PUBLICATIONS

Weighted Trace-Penalty Minimization for Full Configuration Interaction. W. Gao, Y. Li, **H. Shen**. *SIAM Journal on Scientific Computing, 2024*

Resource Efficient Boolean Function Solver on Quantum Computer. X. Li, **H. Shen**, W. Gao, Y. Li. *Quantum, 2024*

Short-term Forecasting of Electric Load Based on Kalman Filter with Elastic Net Method. Y. J. Tian, **H. X. Shen**, W. G. Gao. *IOP Conf. Series: Earth and Environmental Science, 2019*